

INVITED REVIEW

Clinical trials in menopause

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Abstract

Objective: Clinical trials in menopause have undergone much scrutiny over the years. This has led to significant shifts in the treatment of symptomatic menopause and a substantial impact on women. We aim to delineate the key studies contributing to this controversy and highlight new directions specifically related to menopausal hormone therapy (HT) and vascular disease risk.

Methods: We performed a search of sentinel studies delineating the risks and benefits of HT in otherwise healthy postmenopausal women. Using PubMed we input the following search terms: hormone replacement therapy, cardiovascular disease, coronary artery disease, coronary atherosclerosis, myocardial infarction, angina, coronary heart calcification, carotid intimal thickness, lipids, and/or lipoproteins. We included studies of menopausal women (surgical or natural) using combined estrogen/progestogen therapy or estrogen-only therapy that looked at cardiovascular disease risk factors or outcomes. Studies were evaluated for inclusion by the authors; however, this is not intended to be a systematic or an exhaustive analysis.

Results: In women close to the time of menopause, there is a decreased risk of subclinical and clinical coronary heart disease with menopausal HT. Additionally, HT confers a significant benefit to vasomotor symptoms of menopause, bone health, and colorectal cancer. There is an increased risk of venous thromboembolism with oral formulations that appears mitigated with transdermal estradiol. Mixed data regarding breast cancer risk are available, with some studies suggesting an increased risk of invasive breast cancer with estrogen/progestogen therapy and a null effect with estrogen-only therapy. Other more long-term epidemiologic studies identify a decreased risk.

Conclusions: The available literature suggests that HT is a viable option for the primary prevention of cardiovascular disease in postmenopausal women. Newer trials will likely verify this assessment. If this is enough to change clinical practice, however, remains to be seen given the general fear of HT by many with prescriptive authority, and also the women in our care.

Key Words: Cardiovascular disease – Clinical trials – Menopausal hormone therapy.

The history of clinical trials in menopausal hormone therapy (HT) is full of twists, turns, and circles. Data from these trials have been analyzed, re-analyzed,

and interpreted in myriad ways. In this review, we aim to distill this information and make suggestions based on the most up-to-date evidence specifically regarding menopausal

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HT and prevention of cardiovascular disease (CVD). Heart disease is the leading cause of death in women.¹ Weight loss, exercise, and smoking cessation can decrease CVD risk by 12% to 14%.² However, the commonly used pharmacologic preventative strategies used in men, aspirin and statin therapy, do not consistently improve outcomes in women.³ Additionally, whereas women have an overall lower incidence of CVD, they fare worse after a cardiovascular event than men.⁴ Historically, estrogen has been lauded as cardioprotective and has been linked to the improved incidence of CVD seen in women compared with men.⁵⁻⁸ To support this, many observational studies found HT decreases the risk of coronary heart disease (CHD) in the populations observed.⁹⁻¹¹ However, in 2002, the estrogen/progestin arm of the Women's Health Initiative (WHI) was prematurely closed due to concerns for increased risks of both heart disease and breast cancer in women on combined estrogen/progestogen therapy (EPT). An increased stroke risk in women on estrogen-only therapy (ET), along with "futility" or lack of benefit, led to the early termination of that study 2 years later.^{12,13} Upon further analysis of the WHI data and with support of newer studies, the pendulum is swinging again.¹⁴⁻¹⁹ Ultimately, we are left with the question, "Do clinical trials in menopause indicate HT should be used for the primary prevention of CVD in postmenopausal women?"

METHODS

We performed a search of sentinel studies delineating the risks and benefits of HT in otherwise healthy postmenopausal women. Using PubMed, we input the following search terms: hormone therapy (HT), coronary vascular disease, coronary artery disease (CAD), coronary atherosclerosis, myocardial infarction, angina, coronary heart calcification, carotid intimal thickness, lipids, and/or lipoproteins. We included studies of menopausal women (surgical or natural) using HT (EPT or ET) that looked at CVD risk factors or outcomes. Studies were evaluated for inclusion by both authors, but this is not intended to be a systematic or an exhaustive analysis.

BACKGROUND

Since Premarin was approved for the treatment of symptoms in menopause in 1942, we have been dissecting the risks and benefits of HT. In observational trials, HT has been shown to decrease risks of CHD and mortality.⁹⁻¹¹ There were calls to consider HT in all menopausal women, despite symptom status, although most cited a need for randomized trials to confirm these findings. The WHI was meant to be this trial and many papers looked forward to the results with optimism. Then, the WHI was ended prematurely in 2002, citing concerns for increased risks of breast cancer (a prespecified stopping boundary), and CHD in women on EPT.¹² The estrogen-only arm was stopped soon thereafter (in 2004), citing increased risk of stroke and anticipated lack of CHD benefit.¹³ A seismic shift occurred as women stopped taking and physicians stopped prescribing HT.²⁰ For the next 10 years, the WHI data were analyzed and re-analyzed.¹⁴⁻¹⁶ This led to several papers suggesting the risks in

women close to the time of menopause were overstated and the timing hypothesis was born.^{19,21-23} The timing hypothesis posits that when HT is initiated close to the time of menopause, it seems to be cardioprotective. Yet, when initiated later, there is increased risk of CVD and cardiovascular events. Women in the WHI were, on average, 63 years of age at the time of enrollment in the trial. Perhaps, this contributed to the differences in results compared with earlier observational studies involving younger women. Indeed, on further analysis of the WHI participants, in the subset of women 50 to 59 years of age, the risk of CHD was reduced.^{16,24} These results are consistent with animal and bench studies looking at the effects of estrogen on coronary artery plaques.²⁵⁻²⁷ Estrogen is protective against plaque formation, but has a destabilizing effect on plaques that have already formed. Newer trials (KEEPS, ELITE; see below) have been designed to answer the question of HT initiation in younger women and the timing hypothesis. One has shown a null effect in very low-risk women within 4 years of menopause, whereas the other shows promise to confirm this hypothesis and may change the narrative once again.^{17,19}

THE MENOPAUSAL TRANSITION AND CARDIOVASCULAR HEALTH

The menopausal transition is a time when women's cardiovascular risk changes significantly and the incidence of CHD events increases substantially. Early menopause (<50 years of age) is associated with increased CVD events.²⁸ This transition was further illuminated in the Study of Women's Health Across the Nation (SWAN). SWAN was a prospective study designed to characterize the menopausal transition amongst a diverse group of American women. Investigators recruited 3,302 women between 42 and 52 years of age at seven US sites from 1995 to 1997.²⁹ Reports from the SWAN cohort have delineated the physiologic changes that occur with menopause that may lead to cardiovascular risk, including increasing triglycerides, low-density lipoprotein (LDL)-cholesterol, apolipoprotein (B), and other proatherogenic changes.^{30,31} Therefore, perimenopause and early menopause is an ideal time to target the cardiovascular health of women.

EARLY OBSERVATIONAL STUDIES

As stated, many early observational studies noted HT had a significant protective effect on the heart. Initial studies focused on the lipid changes associated with HT including a significant increase in high-density lipoprotein (HDL)-cholesterol (on average 10%) and decrease in LDL-cholesterol (on average 4%).³² Other beneficial lipid changes that likely contribute to the decreased cardiovascular risk noted with HT include decreases in lipoprotein (a) and apolipoprotein B, and also increases in apolipoprotein A1 concentrations.^{33,34} However, these changes cannot account for all of the cardiovascular benefits of HT, as they were also associated with an increase in triglycerides thought to be an independent risk factor for CHD in women. After statistical adjustment for these lipid changes, ET seems to remain a statistically significant predictor of reduced risk of death due

to CVD.³⁵ These results were not isolated as other investigators in these early observations also found a decrease in mortality of up to 41% with moderate estrogen use in menopause.³⁶

Other large cohort studies examining cardiovascular health performed subgroup analyses on women taking HT. These included the Cardiovascular Health Study (CHS), Atherosclerosis Risk In Communities (ARIC) Study, and the Rotterdam Study.³⁷⁻³⁹ In the CHS, approximately 12% of the 2,955 women studied were currently taking ET and an additional 26.5% had used ET in the past. Estrogen use (past or present) was strongly associated with several favorable cardiovascular risk factors including lower LDL-cholesterol, fibrinogen, glucose, insulin, and bodyweight, and higher HDL-cholesterol (all $P < 0.0001$).³⁷ Estrogen users also had lower levels of subclinical disease as measured by carotid intima medial thickness (CIMT) ($P < 0.02$). McGrath et al⁴⁰ duplicated these results several years later. These investigators used CIMT to identify the presence and progression of subclinical atherosclerosis. CIMT is used in many trials of cardiovascular health as one of earliest detectable changes in the development and progression of atherosclerosis and is predictive of clinical cardiovascular events.⁴¹⁻⁴⁵ In ARIC, women on HT ($n = 5,436$) were found to have favorable lipid and hemostasis parameters, but there was no difference in CIMT.³⁸ Women ($n = 1,103$) who underwent nonsurgical menopause between ages 55 and 80 years were included in the Rotterdam Study.³⁹ Ever use of HT was associated with a decreased mean CIMT compared with never-users.³⁹

One of the larger cohorts was the Framingham Heart Study. After 24 years of follow-up, Gordon et al⁴⁶ reported on the risk of HT and heart disease in 2,873 women. They noted HT users had a two times higher risk of CHD; however, this was not statistically significant when angina was omitted as an outcome measure. There was no difference in total mortality.⁴⁶ Later, Wilson et al⁴⁷ described a 50% elevated risk of cardiovascular morbidity ($P < 0.01$) with HT use and, again, no difference in mortality.

At approximately the same time, results from the Framingham Study were being reported, recruitment for one of the largest cohort studies, the Nurse's Health Study (NHS), began.⁴⁸ The NHS included 121,964 female health workers aged 30 to 55 years at the time of recruitment in 1976. Data on CHD and the use of HT were collected starting in 1978. One of the first publications addressing CHD looked at 31,317 women without a history of coronary disease.⁵ Only 90 women experienced a nonfatal myocardial infarction (MI) or coronary death in the 4-year follow-up period. The relative risk (RR) of CHD with ever-use of HT was 0.5 (95% confidence interval [CI] 0.3-0.8, $P = 0.007$), and RR in current users was 0.3 (95% CI 0.2-0.6, $P = 0.001$).⁵ On further analysis, women with surgical menopause had increased risk of CHD if they did not take HT (RR 2.2, 95% CI 1.2-4.2) and utilization of HT eliminated this risk (RR 0.9, 95% CI 0.6-1.6).⁴⁹ In 1991, Stampfer et al⁶ performed a 10-year analysis on a larger cohort of 48,470 women from the NHS. All of these women had no history of CHD or cancer. There were

224 strokes, 405 major CHD events, and 1,263 deaths from all causes. The RR of major CHD in women currently taking estrogen was 0.56 (95% CI 0.40-0.80). The RR for current and former users of estrogen as compared with never-users was 0.89 (95% CI 0.78-1.00) for total mortality and 0.72 (95% CI 0.55-0.95) for mortality from CVD. Grodstein et al⁸ published further support for the protective effects of estrogen in 1997. It suggested that after 15 years of follow-up, the mortality benefit with HT was significant (RR 0.63, 95% CI 0.56-0.70), although this was attenuated with long-term use (RR 0.80, 95% CI 0.67-0.96). Death from CHD was also markedly decreased with HT use (RR 0.47, 95% CI 0.32-0.69).⁸

Additional support for the NHS results came in 1992 from Sweden.⁷ In a cohort of 23,174 women prescribed noncontraceptive HT observed over a mean of 5.8 years, 227 cases of MI occurred compared with the 281 expected cases (RR 0.81, 95% CI 0.71-0.92). If women were < 60 years of age at entry, they had an even greater reduction in risk of 30% with ET alone (RR 0.69, 95% CI 0.54-0.86) and nearly 50% with EPT (RR 0.53, 95% CI 0.30-0.87). Interestingly, the risk estimates were near unity in the first year of therapy, but declined afterwards.⁷

It is unclear why Framingham showed a null effect (after excluding angina in the final analysis) and the NHS had relatively divergent findings, although it is likely due to the ability to accurately diagnose CHD events, tracking use of HT (dosing, duration), and selection bias as to who is prescribed HT.

EARLY RANDOMIZED CONTROLLED TRIALS

With the observational trials overwhelmingly demonstrating a benefit to women's cardiovascular health, several investigators attempted to confirm these results with randomized trials.⁵⁰⁻⁵⁸ One of the first was the Asymptomatic Carotid Atherosclerosis Progression Study (ACAPS).^{50,59} This study identified 186 women without symptoms of CVD and assigned them to either lovastatin or placebo. Approximately 34% were concomitant users of ET. Users of ET were on average younger than nonusers by 3 years, had higher HDL-cholesterol, lower LDL-cholesterol, and were more likely to have had a hysterectomy. In ACAPS, CIMT tended to progress among ET nonusers, but to regress among ET users ($P = 0.05$).⁵⁹ Lovastatin had a marked impact on CIMT progression ($P = 0.004$) in both groups and reduced LDL-cholesterol by 28%. ET appeared to have little additional effect on CIMT in women assigned to lovastatin.⁵⁹

In 1996, Speroff et al⁵¹ published the Continuous Hormones as Replacement Therapy (CHART) trial data. They studied 1,265 women who were > 40 years of age and < 5 years from spontaneous menopause with an intact uterus. Participants received ethinyl estradiol (EE) with or without norethindrone or placebo. There was increased bone mineral density in the EE groups, and also favorable lipid changes. However, there were unacceptable rates of endometrial hyperplasia in the unopposed EE groups. They concluded that estrogen therapy is beneficial for bone and cardiovascular

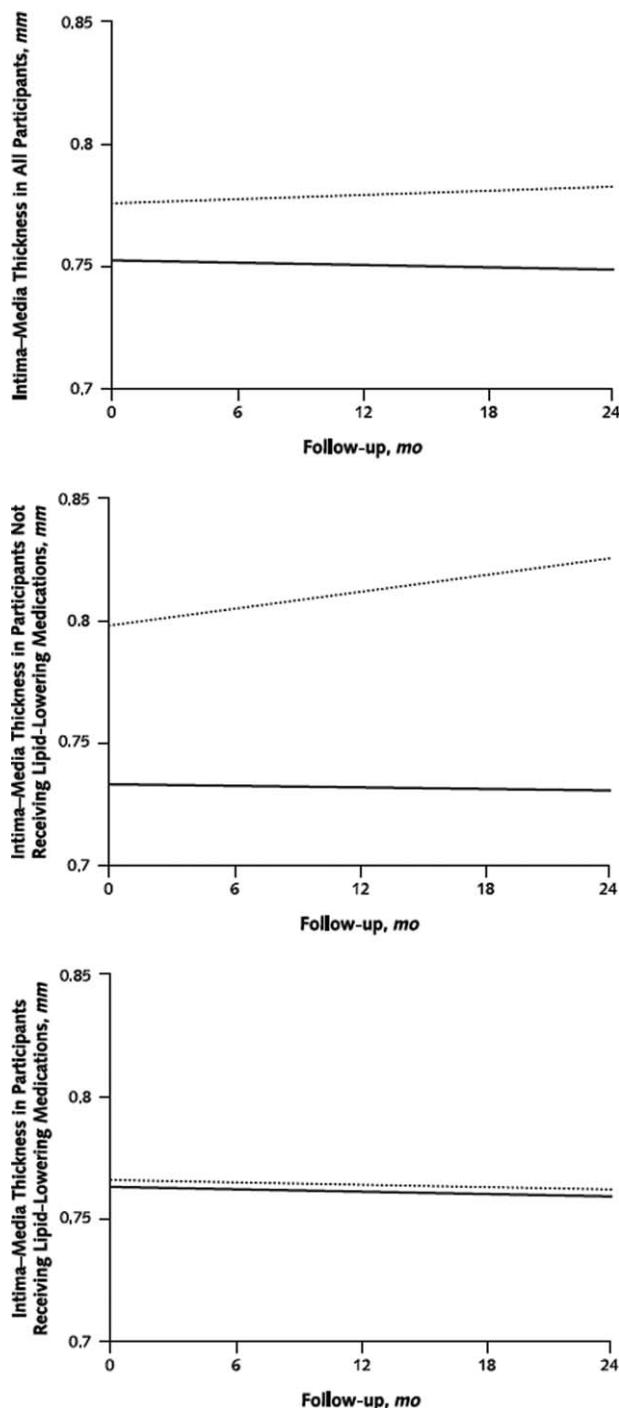


FIG. 1. Time course of progression of common carotid artery intima-media thickness in the estradiol (solid lines) and placebo (dotted lines) groups. Top panel: Stratified by treatment group. The mean ($\pm$ SD) baseline intima-media thickness in all 199 evaluable participants (102 placebo recipients and 97 estradiol recipients) was 0.776 ± 0.149 compared with 0.752 ± 0.111 mm, respectively ($P > 0.2$). The mean rate of change in intima-media thickness in the placebo and estradiol groups was 0.0036 versus -0.0017 mm/y, respectively ($P = 0.046$). Middle panel: Participants not taking lipid-lowering medications. The mean ($\pm$ SD) baseline intima-media thickness in 77 evaluable participants (35 placebo recipients and 42 estradiol recipients) not taking lipid-lowering medications was 0.799 ± 0.182 compared with 0.734 ± 0.111 mm, respectively ($P = 0.07$). The mean rate of change in intima-media thickness in the placebo and estradiol groups was 0.0134 versus -0.0013 mm/y,

health, but should be used with concomitant progestogen therapy in women with an intact uterus, which is now common practice.

At approximately the same time, the Postmenopausal Estrogen/Progestin Interventions (PEPI) trial investigators began to report their findings. The PEPI trial was a randomized controlled trial of 875 women between 45 and 64 years of age who received conjugated equine estrogens (CEE), CEE with medroxyprogesterone acetate (MPA), CEE with micronized progesterone (MP), or CEE with placebo. The baseline characteristics of the cohort were described in 1995.⁶⁰ The group reported favorable lipid profiles in women in all the treatment groups with increased HDL-cholesterol and decreased LDL-cholesterol.⁶¹ Subtle differences in glucose and insulin metabolism, and also weight gain were noted between the MPA and MP groups, favoring MP; however, the clinical significance of these changes remain unknown. Subsequent analysis of adherent participants found similar lipid changes to the intention-to-treat cohort with no significant changes in blood pressure.⁶² Additionally, all treatment groups were found to have decreased lipoprotein (a), with a rapid increase in C-reactive protein (CRP), upon initiation of therapy.^{63,64} This initial increase in CRP has been confirmed by other investigators and is a nonspecific marker of inflammation.⁶⁵⁻⁶⁷ Whereas it is still unclear what impact this CRP phenomenon has clinically, some studies have noted either a neutral or increased risk of CVD in the first year of therapy that may be related.^{7,68-70}

At this point, the idea that HT benefits women's risk of atherosclerosis had gained momentum, and the Estrogen in the Prevention of Atherosclerosis Trial (EPAT) proposed unopposed ET significantly reduces the progression of subclinical atherosclerosis.⁵³ The investigators studied 222 healthy postmenopausal women with an average age of 62.2 years over a 2-year time span. Participants were randomized to receive either estradiol (E_2) or placebo, and measurements of CIMT were obtained. As hypothesized, the E_2 group was noted to have a regression of their subclinical atherosclerosis with a mean change of -0.0017 mm/y versus 0.0036 mm/y in the placebo group ($P = 0.046$).⁵³ An even more profound difference was found in women not taking lipid-lowering medications ($P = 0.002$), whereas in women taking lipid control medications (approximately 62% of the study group), it had a null effect (Fig. 1).⁵³ EPAT also concluded that the lipid changes associated with HT only accounted for 30% of the positive effect on atherosclerosis progression.⁷¹ The remainder of the benefit is likely due to other factors influencing cardiovascular risk including increased estrogen and sex hormone-binding globulin.⁷²

respectively ($P = 0.002$). Bottom panel: Participants taking lipid-lowering medications. The mean ($\pm$ SD) baseline intima-media thickness in 122 evaluable participants (67 placebo recipients and 55 estradiol recipients) taking lipid-lowering medications was 0.764 ± 0.128 compared with 0.766 ± 0.110 mm, respectively ($P > 0.2$). The mean rate of change in intima-media thickness in the placebo and estradiol groups was -0.0016 versus -0.0019 mm/y, respectively ($P > 0.2$). Reproduced with permission from Hodis et al.⁵³

The Women's International Study of Long Duration Oestrogen after Menopause (WISDOM) showed promise to answer many outstanding questions about the long-term risks and benefits of HT.⁵⁶ Postmenopausal women ($n=5,694$) between 50 and 69 years of age were randomized to CEE alone, CEE with MPA, or placebo. Initial findings demonstrated that HT started many years after menopause (on average >15 years) increased the risk of CHD (7 vs 0; $P=0.016$) and venous thromboembolism (VTE) (22 vs 3; hazard ratio [HR] 7.36, 95% CI 2.2-24.6).⁷³ There was no difference in other clinical outcomes including breast cancer, stroke, fracture, and overall death. Given the average age of participants was 62.8 years, it is challenging to extrapolate these data to a younger cohort of women. However, it does suggest initiating HT in these older women confers increased risk. Unfortunately, the study was stopped early due to the release of the WHI findings in 2002 after recruiting only 26% of its target sample and a mean follow-up of 11.9 months.⁷⁴

This leads us to the clinical trial that arguably has had the most significant impact in menopause care—the WHI.

THE WOMEN'S HEALTH INITIATIVE

The WHI was both a randomized controlled and observational trial designed to study various health concerns affecting women. The HT arm of the trial hypothesized HT would reduce the risk of CHD and other CVD risk factors, and, secondarily, reduce the risk of hip and other fractures, while understanding an increased risk of breast cancer was a possible adverse outcome. The trial recruited 64,500 women in the randomized controlled trial and 100,000 women to the observational trial between the ages of 50 and 79 years (mean 63.3 years) in the years 1993 to 1998. Some of these women ($n=27,500$) were included in the HT clinical trials.^{12,13} Women were randomized to either CEE with MPA versus placebo if they had an intact uterus ($n=16,608$) or CEE alone versus placebo if they had a hysterectomy. Primary endpoints were CHD and invasive breast cancer, with a global index (that had not been previously validated) including stroke, pulmonary embolus, endometrial cancer, colon cancer, hip fracture, and death due to other causes as a secondary endpoint. These studies were highly anticipated by many to conclusively determine that HT (CEE with MPA) and ET (CEE alone) reduced the risk of CVD and, as such, could be used for primary prevention.

In June of 2002, the investigators announced the closure of the EPT arm after a mean follow-up of 5.2 years due to increased risks of breast cancer, CHD, stroke, and VTE in women taking HT compared with placebo.¹² Whereas there was a decrease in the risk of hip fracture and colon cancer, the increased risks were deemed to be too significant to continue the trial. Specifically, the HR was 1.29 (95% CI 1.02-1.63) for CHD, 1.26 (95% CI 1.00-1.59) for breast cancer, and 1.41 (95% CI 1.07-1.85) for stroke (Fig. 2). Then, in 2004, the ET arm, comprised of 10,739 women, was closed after a mean follow-up of 6.8 years due to an increased risk of stroke (12 additional per 10,000 woman-years).¹³ A null effect on CHD

was noted, and the investigators determined it would be futile to continue the trial in the face of the increased risk of stroke.

These findings were surprising at the time and led to significant media coverage and a shift in women's attitudes and usage of HT.^{20,75-77} The WISDOM investigators surveyed participants in 2004 to better understand women's perceptions of HT after the WHI findings were released. In one study, women seemed to be aware of an increased risk of breast cancer and decreased risk of hip fracture associated with HT. However, they seemed less convinced regarding dementia and CVD risk.⁷⁵ Many were willing to participate in another trial (85%) and most past HT users resumed their therapy. Another study found that many women believed that the decision to continue HT was a personal one and quality-of-life factors played a key role in this decision.⁷⁶ However, a year later, another report found that many women had heard of the WHI findings and 52% reported that it affected their use of HT and they were significantly less likely to trust their physicians' recommendations regarding HT.⁷⁸ This trend was also reflected in prescribing trends, with HT claims reduced by more than 70% in the 7-year period after the WHI.²⁰

The WHI data have since been dissected further with the hopes of determining why their findings were so disparate from earlier trials.^{14-16,22,24,79} Much of the difference was thought to be due to the age of the WHI participants in comparison with earlier trials. Many women in the WHI were >10 years from the menopausal transition and were, on average, older than the women represented in other studies. Only 16.5% of women in the EPT arm of the study were within 5 years of menopause and 574 women were 50 to 54 years of age on recruitment into the WHI. Indeed, when CHD with EPT was explored further, Rossouw et al²⁴ found the risk was null until a woman was ≥ 20 years from the menopausal transition, with a HR of 0.76 (95% CI 0.5-1.16) in the first 10 years after menopause (P for the trend = 0.02). This corresponds to an estimated absolute risk of -6 per 10,000 women-years in this same time period. Similarly, in women 50 to 59 years of age, the HR for CHD was 0.93 (95% CI 0.65-1.33). Younger women also had a nonsignificant

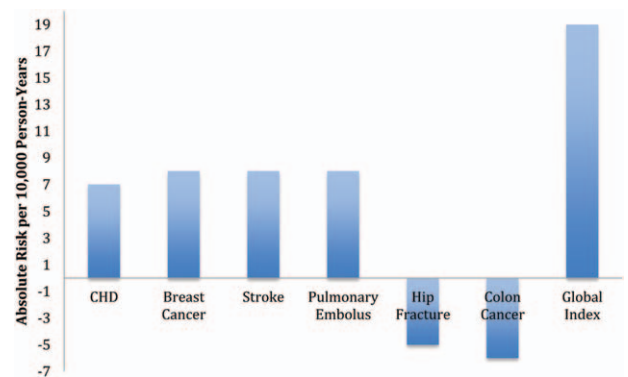


FIG. 2. Absolute risk per 10,000 person-years attributable to estrogen plus progestin therapy according to Women's Health Initiative results (Rossouw et al¹²). CHD, coronary heart disease.

trend toward decreased total mortality with HT.²⁴ Additional data on this cohort of younger women were followed, and, after an average of 7.4 years of follow-up, women on CEE were found to have significantly reduced progression of coronary artery calcification, and also CHD risk (Fig. 3).^{14,80} This ET arm was followed for another 3 years, with a null effect on nonfatal MI and fatal CHD regardless of age, and decreased total mortality and global index in women 50 to 59 years of age.¹⁵ Ultimately, after 13 years follow-up, women 50 to 59 years of age taking CEE alone had reduced all-cause mortality, MI (by 40%), and improved global index. Absolute risks for women 50 to 59 years of age were low and significantly less than older women.¹⁶ Perhaps, it is this timing hypothesis that explains the differences between the WHI and other trials.

THE TIMING HYPOTHESIS

The “Timing Hypothesis” (aka the “Gap Hypothesis”) implies that initiation of HT close to the time of menopause has CVD benefit compared with later initiation. As we have discussed, further analysis of the WHI data seems to support an either null or decreased risk in CVD in women <10 years from menopause or in the 50 to 59-year age group.^{14-16,24} This hypothesis is supported by several interesting studies on the development and progression of CVD.²⁵⁻²⁷ In mice, estrogen appears to inhibit early formation of coronary plaques via inhibition of the fatty streak.²⁵ However, intraplaque hemorrhage and progression of established lesions is unaltered. This phenomenon has also been demonstrated in monkeys where estrogen given close to surgical menopause decreases atherosclerosis by approximately 70%.²⁶ This same benefit was not seen in late surgically menopausal monkeys who showed increased plaque instability. Additionally, estrogen has been shown to be a key factor in arterial collateral growth after arterial occlusion.²⁷

In humans, data from the NHS showed that HT initiated <4 years from the time of menopause was associated with a reduced risk of CHD with ET (RR 0.66, 95% CI 0.54-0.80) or EPT (RR 0.72, 95% CI 0.56-0.92). In women >10 years from menopause, the risk reduction was not significant.⁸¹ Interestingly, Toh et al²³ found a small increased risk of CHD with HT use in the first 2 years after the menopausal transition (HR 2.36, 95% CI 1.55-3.62).²³ This was followed by a null effect (HR 1.69, 95% CI 0.98-2.89). Early observational trials potentially missed this increased risk and may only reflect the later effect, helping us to further reconcile the variance with the WHI.

META-ANALYSES

Several meta-analyses have attempted to put this somewhat conflicting data together. Before the WHI, many hospital-based case-control studies showed an increased risk of CVD with HT use. However, other studies showed improvement, including population-based case-control studies, cross-sectional analyses, and prospective studies. A weighted precision-based analysis of available quantitative data in

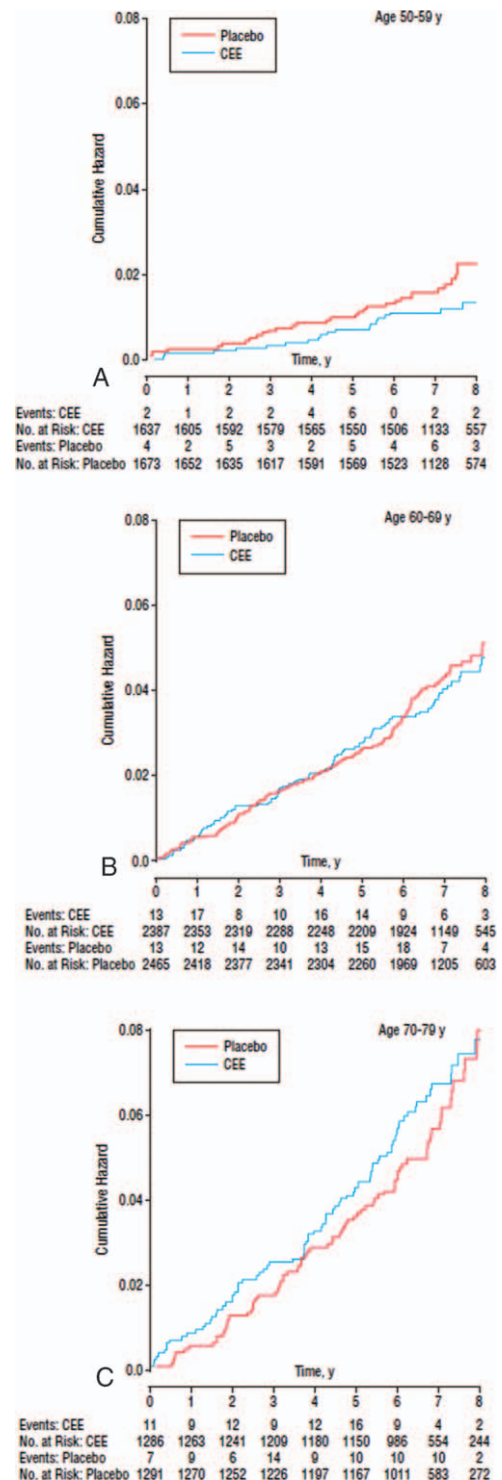


FIG. 3. Kaplan-Meier estimates of cumulative hazard ratios for coronary heart disease (myocardial infarction or coronary death) by age decade according to Women's Health Initiative results for women receiving conjugated equine estrogen (CEE) or placebo: (A) 50-59, (B) 60-69, and (C) 70-79 years (reproduced with permission from Hsia et al⁷⁹).

1991 revealed a RR of 0.56 (95% CI 0.5-0.61). When only controlled prospective and angiographic studies were considered, the RR was 0.5 (95% CI 0.43-0.56).⁹ These findings were supported by two other meta-analyses, using pooled data

to create summary RRs that showed additional benefit regarding hip fracture (RR 0.75, 95% CI 0.68-0.84) and increased risk of breast cancer (RR 1.25, 95% CI 1.04-1.51).^{10,11}

After the WHI, Salpeter et al⁸²⁻⁸⁴ performed a series of meta-analyses of randomized trials that included the WHI data. In 2004, 30 trials including 26,708 women were analyzed and they found the pooled odds ratio (OR) for total mortality with HT showed a null effect at 0.98 (95% CI 0.87-1.12). As additional support for the timing hypothesis, mortality in younger women was significantly reduced (OR 0.61, 95% CI 0.39-0.95), but not in older women (OR 1.03, 95% CI 0.90-1.18).⁸² A subsequent analysis of studies looking at pooled ORs for CHD endpoints as opposed to mortality alone revealed HT significantly reduced CHD events in younger women with an OR of 0.68 (95% CI 0.48-0.96). Older women showed a null effect (OR 1.03, 95% CI 0.91-1.16).⁸³ This meta-analysis also demonstrated the increase in CHD events in the first year of use seen by Toh et al.⁷⁰ Further evidence, using a hierarchical random-effects Bayesian meta-analysis of mortality risk in younger women (average age 55 years), included age-specific data from the WHI and showed an approximately 30% reduction with the use of HT (RR 0.73, 95% CI 0.52-0.96).⁸⁴

In 2012, a Cochrane review looked at 23 studies including 42,830 women.⁸⁵ Approximately 70% of the data was extracted from the WHI and The Heart and Estrogen/Progestin Replacement Study (HERS) that looked at secondary prevention of CVD with HT. The mean age of the participants was >60 years. This analysis showed that women taking EPT had increased risk of coronary events, VTE, breast cancer, and stroke. ET significantly increased the risk of VTE and stroke, but did not significantly increase the risk of breast cancer. They concluded that HT should not be used for the primary or secondary prevention of CVD in women.⁸⁵ A Cochrane update in 2015 included an additional six studies totaling 40,410 women. The entire cohort demonstrated little benefit with regards to mortality or CVD, but did have an increased risk of VTE and stroke with a high level of evidence.⁸⁶ A subgroup analysis of women initiating HT within 10 years of menopause showed a lower mortality (RR 0.70, 95% CI 0.52-0.95) and risk of CVD (RR 0.52, 95% CI 0.29-0.96).⁸⁶ However, these women were still at increased risk of VTE. They also concluded that HT should not be used for the primary or secondary prevention of CVD.

AFTER THE WHI

With results from the WHI, meta-analyses, and Cochrane reviews, women and providers were reeling. Some evidence showed a significant benefit to HT, whereas some showed a significant harm. There were calls for further studies and analyses: several ongoing and also new studies emerged. In 2008, the intention-to-treat analysis from the NHS findings was published. This demonstrated a HR of 0.54 (95% CI 0.19-1.51) in women within 10 years after menopause and 1.2 (95% CI 0.78-1.84) in others once adjusted for adherence.⁶⁹

Whereas these data did not show a significant benefit of HT, it did not show an increased harm that was the critical finding in the wake of the WHI. The Women Ischemia Syndrome Evaluation (WISE) trial was first described in 1999.⁸⁷ The investigators sought to optimize cardiac care for women by optimizing symptom evaluation and diagnostic testing, and also evaluate the influence of reproductive hormones. They recruited 936 women who were undergoing a cardiac catheterization for nonocclusive disease. The average age of participants was 59 (range 21-86) years and 86% of the cohort was menopausal. Recruitment was closed early after the WHI findings. After a 6-year follow-up, approximately 25% of the cohort had obstructive CAD. This group was more likely to be diabetic, older, not using HT, and using lipid-lowering medication. Additionally, they found women never using HT had a higher incidence of CAD than women using HT for <5.5 years. Women using HT for >5.5 years had an even lower incidence of CAD than the aforementioned groups.⁸⁸ Further data from 35 participants showed HT improved rates of chest pain, vasomotor symptoms, and quality-of-life measures, but had no impact on MI or endothelial dysfunction.⁸⁹

Two additional studies conceived and started before the WHI that should be mentioned are the Danish Osteoporosis Prevention Study (DOPS) and results from a Finnish cohort followed since 1994.^{58,90-92} In DOPS, investigators hoped to see a reduction in osteoporotic fracture with early initiation of HT in 2,016 healthy postmenopausal women (average age 50 years on enrollment). Secondly, they looked at a composite endpoint of mortality, MI, and heart failure in 1,006 women enrolled in a randomized substudy of CVD risk. In initial reports, women taking HT were found to have improved vasomotor symptoms, vaginal dryness, and bone mineral density.^{93,94} In 2012, the data from the cardiovascular substudy revealed a significantly lower risk of the composite outcome after 10 years of randomized treatment (HR 0.48, 95% CI 0.27-0.89, $P = 0.015$) and 16 years of follow-up (HR 0.61, 95% CI 0.39-0.94) in women who received HT compared with those who did not (Fig. 4). Total mortality did not decrease significantly at either time-point.⁹⁵ They also found no increase in breast cancer ($P = 0.71$) and a nonsignificant increase in DVT and stroke.⁹⁵

In Finland, HT was initially recommended as primary prevention of CVD, given the high rates of vaso-occlusive heart disease seen there, and also being based on data from early observational trials.⁹⁶ HT use was recorded in public registries starting in 1994 and 489,105 women using HT over a total 3.3 million exposure-years were available for analysis. They found the risk of cardiovascular death was significantly reduced by 18% at 1 year of exposure and further reduced by 38% at 3 to 5 years of exposure to HT. All-cause mortality was reduced by 12% to 38%, and the percent increased in relation to exposure time.⁹⁰ When the timing of initiation of HT was studied, age <60 years was found to be related to a significant reduction in cardiovascular mortality, with a mortality ratio of 0.53 (95% CI 0.47-0.59). Initiation at >60 years of age was

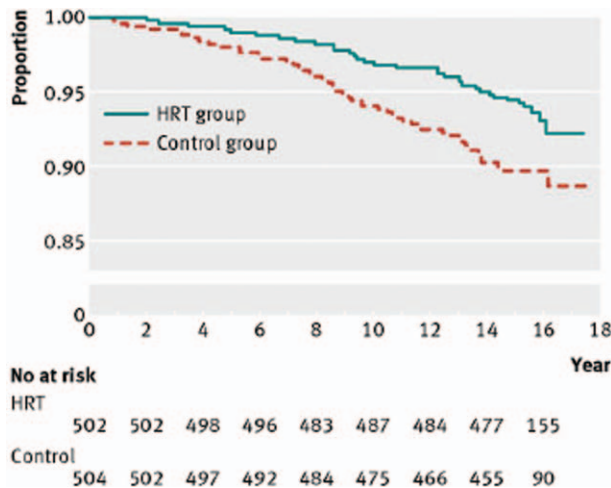


FIG. 4. Risk of death or admission to hospital due to heart failure or myocardial infarction over 16 years of follow-up including 11 years of randomized treatment with hormone therapy (HT) or controls in the Danish Osteoporosis Prevention Study (reproduced with permission from Schierbeck et al⁹⁵). HRT, hormone replacement therapy.

not associated with a reduction or increase in mortality. All hormone users (ET or EPT) showed reduced death risk the earlier HT was started ($P < 0.05$).⁹¹ With the release of the WHI results, concern that widespread discontinuation of HT would lead to worse outcomes was prevalent. Mikkola et al⁹⁷ reported on a cohort of 332,202 women who had discontinued HT between 1994 and 2009. They found the risk of cardiac death was significantly increased in the first post-treatment year, with a mortality ratio of 1.26 (95% CI 1.16-1.37), whereas there was a reduction in mortality after 1 year. If women discontinued HT at <60 years of age, they had an increased mortality that was not reproduced in women discontinuing at >60 years of age.⁹⁷ Additionally, these results did not differ in the years before (1995-2001) and after (2005-2009) the WHI.⁹²

There have been two main randomized, placebo-controlled clinical trials initiated after the WHI that have further clarified our view of HT and CVD risk. These are the Kronos Early Estrogen Prevention Study (KEEPS) and the Early versus Late Intervention Trial with Estradiol (ELITE) study.^{17-19,98} KEEPS was designed to investigate if low-dose HT initiated early in menopause would reduce subclinical atherosclerosis and to determine if there were meaningful differences by route of estrogen administration (oral versus transdermal). This multicenter, double-blind, randomized placebo-controlled clinical trial recruited 727 women with an average age of 52.7 years who were in generally good health. They were on average 1.4 years postmenopause and randomized to oral CEE (0.45 mg/d, slightly lower than the 0.625 mg/d used in the WHI), transdermal E_2 patches (0.05 mg/d), or placebo. Treatment groups received progesterone for endometrial protection. Investigators measured CIMT as their primary endpoint.⁹⁸ After 4 years, postmenopausal HT did not affect progression of atherosclerosis despite improving lipids.¹⁷

Some were surprised by the null effect, which may have been related to the very-low-risk women selected for the study or the low dose of estrogen used.

The ELITE was designed to further investigate the timing hypothesis. Investigators recruited 643 healthy postmenopausal women from 2005 to 2008. They were divided into two groups: an early group <6 years from menopause (average 55.4 years of age and 3.5 years from menopause) and a late group (average 65.4 years of age and 14.3 years from menopause). The late group was more likely to have used HT in the past and be current users of antihypertensive and lipid-lowering medications. Women were randomized to receive oral E_2 (with or without vaginal progesterone depending on an intact uterus) or placebo for a mean of 6 to 7 years. Baseline results showed the mean CIMT was greater in the late group than the early group, with blood pressure and body mass index positively associated with CIMT in both groups. There was also a significant positive association of CIMT with LDL-cholesterol, and inverse association with HDL-cholesterol and sex hormone-binding globulin.¹⁸ After a mean of 5 years follow-up, HT affected the early group more favorably than the late group ($P = 0.007$ for the interaction). The early group had significantly less progression of CIMT while on HT ($P = 0.008$) than placebo. The effect in the late group was not significant ($P = 0.29$). Additionally, there were no differences in breast cancer rates, MI, or VTE in any of the groups.¹⁹ The ELITE findings are promising and correlate with the preponderance of the evidence that shows HT is likely cardioprotective when initiated close to the time of menopause in otherwise healthy women.

SECONDARY PREVENTION

Given HT has been found to have a significant positive impact on CVD risk in otherwise healthy early menopausal women, many investigators have hoped to find a similar effect in women with established disease. The Estrogen in Women and Atherosclerosis (EWA) study aimed to demonstrate similar lipid changes with HT in 118 women with angiographically proven CAD as compared with healthy women. They found that after 12 months of transdermal estrogen therapy (with or without cyclic progesterone), no additional effect in lipids was noted. However, the majority of these women were treated with statins at the time of the study.⁹⁹ The concomitant use of statins with HT does not seem to have cumulative benefit to lipids.⁵⁹ The Postmenopausal Hormone Replacement on Atherosclerosis (PHOREA) study also looked at subclinical markers of CHD including CIMT and lipids. They found that HT did not slow CIMT progression, but did significantly decrease LDL-cholesterol.⁵⁴

Although HT did not impact risk factors for recurrent CHD events, were clinical outcomes changed? The NHS took 2,489 women with history of CAD or MI and investigated the impact of HT on recurrent MI or cardiovascular death.⁶⁸ They noted a decreasing risk for recurrent CVD events with increasing duration of hormone use (P for trend = 0.002) and

a RR of 0.38 for longer-term users (95% CI 0.22-0.66). After 20 years of follow-up, the RR remained favorable (RR 0.65, 95% CI 0.45-0.95).⁶⁸ Another study with long-term data was the Estrogen Replacement and Atherosclerosis (ERA) study that hypothesized HT would slow the progression or induce regression of coronary artery atherosclerosis when compared with placebo.¹⁰⁰ This randomized controlled trial of 309 women over an average of 3.2 years looked at CVD events and also lipids, lipoproteins, and other hemostatic factors. They noted significant reduction in LDL-cholesterol and significant increase in HDL-cholesterol, but no alterations in the progression of coronary disease or clinical cardiovascular events.¹⁰¹ The Heart and Estrogen/Progestin Replacement Study (HERS) looked at 2,763 women and found no difference in long-term outcomes (RR 0.99, 95% CI 0.8-1.22) with an increased risk in the first year.¹⁰² The Estrogen for the Prevention of Re-Infarction Trial (ESPRIT) confirmed these findings with no differences in re-infarction rate and cardiac death in women with one previous MI taking HT.¹⁰³ In a 14-year follow-up report, the all-cause mortality showed no significant difference between treatment groups.¹⁰⁴ The Women's Angiographic Vitamin and Estrogen (WAVE) trial was one of the few that showed a detrimental rather than a null effect on CHD with HT.¹⁰⁵ Women with an average age of 65 years who had at least a 15% to 75% coronary artery stenosis were found to have a significantly increased risk of a composite outcome including coronary mean luminal diameter, MI, and death ($P = 0.045$).¹⁰⁵ Another was the Papworth HRT Atherosclerosis Study (PHASE) that showed HT might increase CVD events by 30% to 50%, although the P value was not significant.¹⁰⁶

Three studies evaluating the secondary prevention of CHD were published after the WHI EPT results were released in 2002. These include Women's Estrogen/Progestin Lipid Lowering Atherosclerosis Regression Trial (WELL-HART), Estrogen and Graft Atherosclerosis Regression (EAGAR) trial, and Women's Hormone Intervention Secondary Prevention (WHISP) study.¹⁰⁷⁻¹⁰⁹ WELL-HART found no differences in coronary lesion progression among treatment groups and no increases in cardiovascular events.¹⁰⁷ The EAGAR trial investigated 83 women started on HT within 6 months of coronary bypass.¹⁰⁸ Interestingly, HT slowed angiographic progression of graft disease, but accelerated disease progression in nonbypassed native coronary arteries. Extrapolated, we could expect eight more women would require angioplasty if they utilized HT ($P < 0.05$).¹⁰⁸ The WHISP study was a randomized trial designed to see if low-dose HT would not harm women with acute coronary syndrome.¹⁰⁹ They found no significant differences in lipid levels between the two groups, although a significant percentage of their population was using statins at the time of discharge from the study (79.6% in the treatment group and 82.4% in the placebo group, not significantly different between groups). There was no increased risk of adverse cardiovascular outcomes (OR 0.63, 95% CI 0.31-1.31), indicating HT may be given safely to women with acute coronary syndrome.¹⁰⁹

Ultimately, HT is likely not the answer, and secondary prevention of CVD events in postmenopausal women should focus on lifestyle modification and the use of other CVD prevention tools.⁶⁵

VASOMOTOR SYMPTOMS

Whereas we have thus far focused on the CVD risk associated with menopausal HT, we should also explore some of the other risks and benefits. The decision to initiate HT is not made in a vacuum. There are myriad reasons women and their providers consider HT. Vasomotor symptoms occur in approximately 75% of postmenopausal women.¹¹⁰ HT is one of the most effective treatments for vasomotor symptoms and can decrease the frequency and severity of hot flashes and night sweats.¹¹¹⁻¹¹³ Analysis of the WHI cohort revealed a significant improvement in hot flashes with HT versus placebo (85.7% vs 57.7%; OR 4.40, 95% CI 3.40-5.71) and night sweats (77.6% vs 57.4%; OR 2.58, 95% CI 2.04-3.26).¹¹⁴ The DOPS investigators found similar improvement in hot flashes ($P < 0.01$) and night-time vasomotor symptoms ($P < 0.01$).⁹³

VENOUS THROMBOEMBOLISM

The risk of VTE is consistently increased with oral HT preparations in both observational and randomized trials.¹¹² One meta-analysis of several observational studies showed an OR of 1.57 (95% CI 1.48-1.68) for low-dose HT versus placebo that was increased further with higher doses of estrogen (OR 1.91, 95% CI 1.70-2.16).¹¹⁵ The WHI data revealed a two-fold greater risk of VTE with combined EPT versus placebo.¹² In DOPS, there was a three-fold greater risk of VTE reserved to the first year of use for the entire cohort.¹¹⁶ Additionally, they found younger postmenopausal women did not have a significantly increased risk of VTE ($P = 0.29$). The Estrogen in Venous ThromboEmbolism Trial (EVTET) found that VTE in women with a prior history was also increased in HT users compared with matched controls (10.7% vs 2.3%). This trial was stopped early due to the above results.¹¹⁷

Transdermal preparations do not seem to confer the same risk as oral estrogen treatments.^{118,119} The ESTrogen and THromboEmbolism Risk (ESTHER) study was a French case-control study that investigated the risk of VTE with oral versus transdermal estrogens.^{118,120} They found no increased risk with transdermal preparations compared with nonusers, with an OR of 0.9 (95% CI 0.4-2.1).¹¹⁸ Another study found the RR of VTE with transdermal estrogen was 1.0 (95% CI 0.9-1.1).¹¹⁹ Additionally, Bergendal et al¹²¹ confirmed that transdermal estrogen did not increase VTE risk with a case-control study in 2016. Transdermal estrogen also seems to have a cardiovascular benefit. In a retrospective cohort of 2,551 first-time users of ET, CVD risk was decreased by 20% with transdermal estrogen compared with oral therapy.¹²² This was due to a decreased incidence of angina and congestive heart failure with similar rates of MI and stroke. There was also a 50% reduction in VTE with transdermal ET and a trend to lower cost.¹²²

STROKE

The risk of stroke associated with HT has also been studied extensively. Examination of the Framingham cohort found a two-fold increased risk of cerebrovascular disease without increased mortality ($P < 0.01$).⁴⁷ The NHS, however, found no increased risk of stroke in HT users (RR 0.07, 95% CI 0.65-1.45).⁹ Additional analysis of women 50 to 54 years of age found an increased risk of stroke with HT, with an attributable risk of approximately two cases per 10,000 women per year taking hormones.²¹ Stroke risk was increased in the WHI-EPT arm with a HR of 1.41 (95% CI 1.07-1.85) and in the ET arm, with an additional 12 strokes per 10,000 person-years (HR 0.39, 95% CI 1.10-1.77).¹² The Women's Estrogen for Stroke Trial (WEST) was designed to examine this question further.¹²³ Investigators found no difference in the combined outcome of stroke or death.¹²⁴ More recently, Mikkola et al⁹⁰ examined a Finnish cohort and found an 18% reduction in stroke death. When trends before and after WHI were followed, there were no differences in the incidence of fatal stroke.⁹⁶

BREAST CANCER

An elevated risk of breast cancer is one of the primary reasons the WHI ended prematurely.¹² These results had been previously described in the literature. In one meta-analysis, published in 1992, there was a RR of 1.25 for breast cancer (95% CI 1.04-1.51).¹⁰ The NHS found an increased risk of invasive breast cancer, but no increased mortality due to breast cancer (RR 0.76, 95% CI 0.56-1.02).⁸ Ultimately, the WHI found an increased risk of breast cancer after 3 to 5 years of continuous EPT use.^{12,125} When results from the WHI observational trial and the EPT clinical trial were combined, there was an increased risk for breast cancer when HT was initiated within 5 years of menopause, with an attributable risk of 39% over the first 5 years of use and 54% over the first 10 years of use.¹²⁶ This increased risk was not seen in women who initiated HT >5 years after the menopausal transition. This was an apparent definite shift from the "timing hypothesis" related to CVD and has been confirmed by other investigators.^{127,128}

Additional studies published after the WHI also found increased risk of breast cancer with HT including a 2012 Cochrane review.⁸⁵ In a Finnish cohort of 2,171 women with breast cancer, investigators found E₂ therapy means two to three additional cases of breast cancer per 1,000 women.¹²⁹ This elevated risk was not seen in women with <5 years of HT use. In an updated cohort of 6,211 women with breast cancer, there was no increased risk within the first 3 years of use, but this steadily increased to a RR of 2.07 (95% CI 1.84-2.3) after 10 years of use.¹³⁰

In contrast, several other trials showed either no differences in risk or a reduced risk. In DOPS, the RR of breast cancer was 0.58 (95% CI 0.27-1.27, $P = 0.17$).⁹⁵ Another Danish study showed women using HT had a 47% lower risk of dying from breast cancer compared with never-users.¹³¹ Additionally, the largest cohort to date, encompassing 489,105 women using HT, breast cancer mortality was reduced significantly with a

mortality ratio of 0.56 (95% CI 0.52-0.60) after 5 years of use. This benefit persisted even after >10 years of use.¹³² In younger women (aged 50-59 years), a significantly larger risk reduction was detected (mortality ratio 0.33, 95% CI 0.29-0.37).¹³² ELITE demonstrated no significant differences in breast cancer risk among women randomized to HT or placebo.¹⁹

Estrogen-only therapy alone does not seem to increase breast cancer risk.^{13,85} Furthermore, after 10.7 years of follow-up, there was a decrease in breast cancer in women taking ET, with a HR of 0.77 (95% CI 0.62-0.95).¹⁵ A study observing a large Icelandic cohort of 16,928 women confirmed these results.¹³³

Whether the effects of estrogen with or without progesterone/progestogen on breast cancer risk are related to promotion of or deceleration of current preclinical disease growth, or alternatively on the oncogenic event itself, remain controversial.

Further research regarding HT and breast cancer risk will likely focus on the tissue-specific estrogen/antiestrogen compound—conjugated estrogen and bazedoxifene (CE/BZA). The Selective estrogens, Menopause, And Response to Therapy (SMART) series of clinical trials demonstrated the efficacy and safety of this combination for treatment of menopausal VMS and the prevention of osteoporosis.¹³⁴⁻¹⁴⁶ In the breast, there is evidence of an antiestrogen effect in monkeys.¹⁴⁷ In humans, CE/BZA has demonstrated noninferiority when compared with placebo for mammographic breast density (an independent, moderate risk factor for breast cancer) and breast tenderness (a common reason women discontinue menopausal HT).^{138,142}

COLON CANCER

Colon cancer is the third most common cancer and the third leading cause of cancer death in women.¹⁴⁸ In a meta-analysis of 18 epidemiologic studies, HT use was associated with a 35% (95% CI 17%-50%) lower risk of colon cancer.^{48,149} This benefit was also demonstrated in randomized trials. In the WHI, the risk of colon cancer was significantly reduced in the EPT trial, with a HR of 0.63 (95% CI 0.43-0.92).¹² The ET arm showed a null effect.¹³

FRACTURE

While menopausal HT is not US Food and Drug Administration (FDA)-approved for the treatment of osteoporosis, there have been many studies that have shown its benefit in reducing fractures in postmenopausal women. In the Framingham cohort, the RR of hip fracture in women taking estrogens was 0.65 (95% CI 0.44-0.98).¹⁵⁰ The PEPI investigators showed women in the active treatment groups gained bone density, whereas the placebo groups lost bone density in both the hip and spine.¹⁵¹ WHI demonstrated a significant reduction in hip fracture with both EPT (HR 0.66, 95% CI 0.45-0.98)¹³⁴⁻¹⁴⁴ and ET alone (HR 0.61, 95% CI 0.41-0.91).^{12,13} ET had a favorable effect on bone density in the DOPS population as well.⁹⁴

CONCLUSIONS

Menopausal HT has undergone intense scrutiny. Prescribing practices have changed dramatically over the years and women are increasingly familiar with the data behind these patterns.^{20,73} Nevertheless, perceptions continue to be influenced by the WHI, possibly the most important women's health study of the contemporary era. Unfortunately, this influential study did not enroll enough early menopausal women (approximately 12% aged 50-54 years and 30% aged <60 years) in either its estrogen/progestin¹² or estrogen-alone¹³ substudies to convincingly inform us of the impact of HT in young, early menopausal women. Despite this, the Coordinating Center's and WHI Director's press release ("NHLBI Stops Trial of Estrogen Plus Progestin due to Increased Breast Cancer Risk, Lack of Overall Benefit") confused the prescribing community, and women generally, fueling their fear of breast cancer, and prompting calls for an independent outside review.¹⁵² The tone, analyses conducted, and interpretation of the results in the primary HT paper,¹² along with a lack of transparency and consensus even among the WHI investigators, led to class assumptions, suggesting that all estrogens and progestogens have similar benefits and risks.¹⁵³ Doing so created new problems for all concerned, including "umbrella" class labeling by the US FDA, which was both scientifically unsupported (even by the WHI itself) and detrimental to this day.¹⁵⁴

Many deem the decision to take HT a personal one with a focus on quality of life.⁷⁴ Excellent scientific evidence from over 50 years of both observational studies and clinical trials, many of which are reviewed here, suggests that the benefits of both HT and ET outweigh the risks for most women when started within a few years of their last menstrual period. Ultimately, we can reassure many women that the risks of HT, especially in women 50 to 59 years of age, are low.¹⁵⁵ Additionally, many of these women will likely derive significant benefit not only with regards to menopausal symptoms but also to their hearts. We anticipate another shift in the story to one of primary prevention for all eligible women during the menopausal transition.

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